MINNEMUDAC 2026 · THE FOOD GROUP · GRADUATE TEAM G08

The Food Group can see demand. It can't see need.

Supply follows visits, and visits only count the people who reach a shelf. We set out to answer three questions for Minnesota:

1. Where do need and demand split?
2. Why do they split?
3. What should change on Monday?

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NEED AND DEMAND ARE DIFFERENT NUMBERS · US ADULTS 18–64, DECEMBER 2025



Nearly **3 in 10** adults in food-insecure households needed charitable food and **did not get it**.

Urban Institute, Well-Being and Basic Needs Survey, December 2025 (catalog rows 63–64). National; it frames the question, it does not enter any Minnesota estimate.

Provided data: The Food Group reporting network, January 2022 – July 2026 · Feeding America Map the Meal Gap · BLS food prices · Minnesota DCYF SNAP

Additional public sources: Census ACS and 2020 blocks · USDA LILA, SRAM and RUCC · CDC SVI · BLS LAUS · HUD rents · OpenStreetMap roads · Urban Institute

Open on your phone or tablet

Scan to view this page on your own device.

localhost:3000/present

Need can't be counted directly, so we score the barriers between people and food shelves, in every census tract.

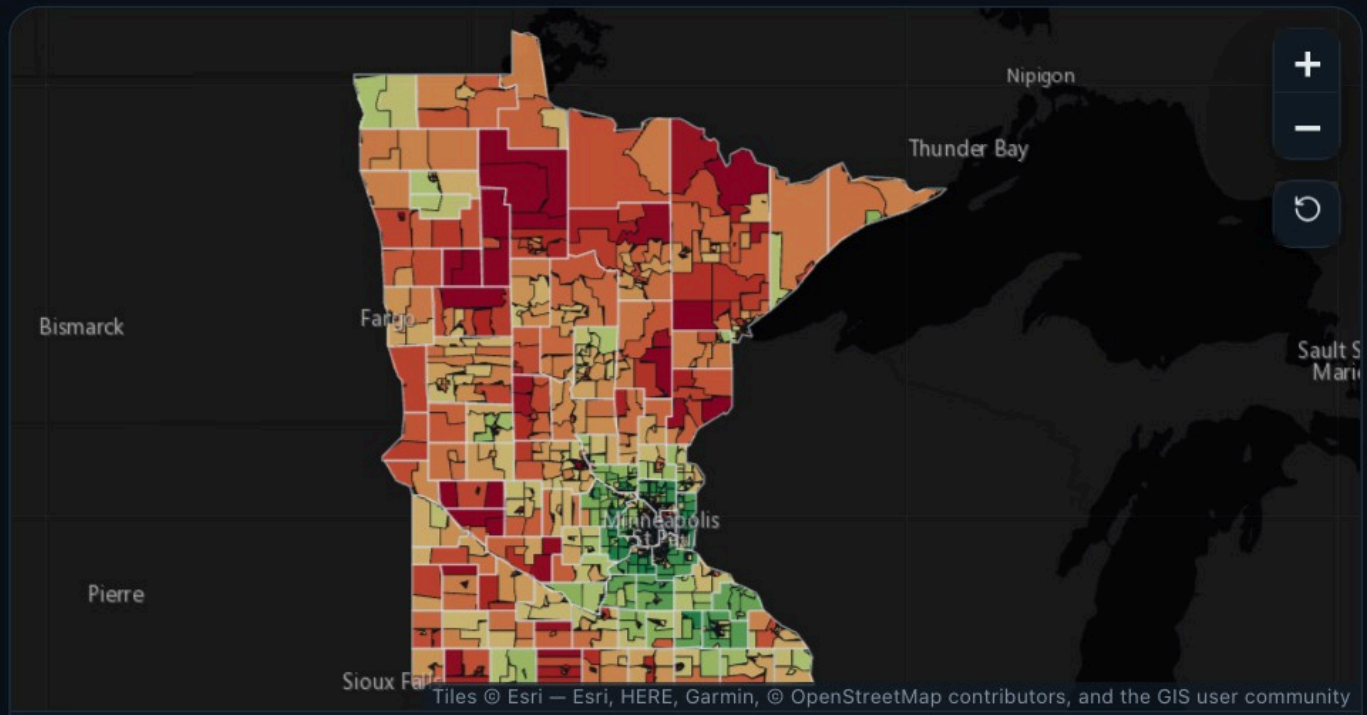
✓ Need · ✓ Supply · ✓ Travel · ✓ Mobility · ✓ Demographics · 6 Need × access = FEI

N	**Need** Money is tight: poverty, rent burden, unemployment	35%
S	**Supply** Little food nearby: pounds within reach per low-income resident	15%
T	**Travel** Far from a shelf: miles to the nearest reporting shelf	20%
M	**Mobility** Hard to get there: households without a car	15%
V	**Demographics** More vulnerable households: children, household make-up	15%

MINNESOTANS BEYOND EASY REACH OF A REPORTING SHELF

2.7M

47% of residents: more than 1 mile away in town, 10 miles in the country.



FEI, 0–100

15 · 29 · 43 · 56 · 70

Fewer barriers — More barriers

Need × access: one 0–100 score per tract, high only where both are high. Red = most barriers.

Monday · Allocation: weigh supply by barriers as well as by visits; visits only show who already gets in.

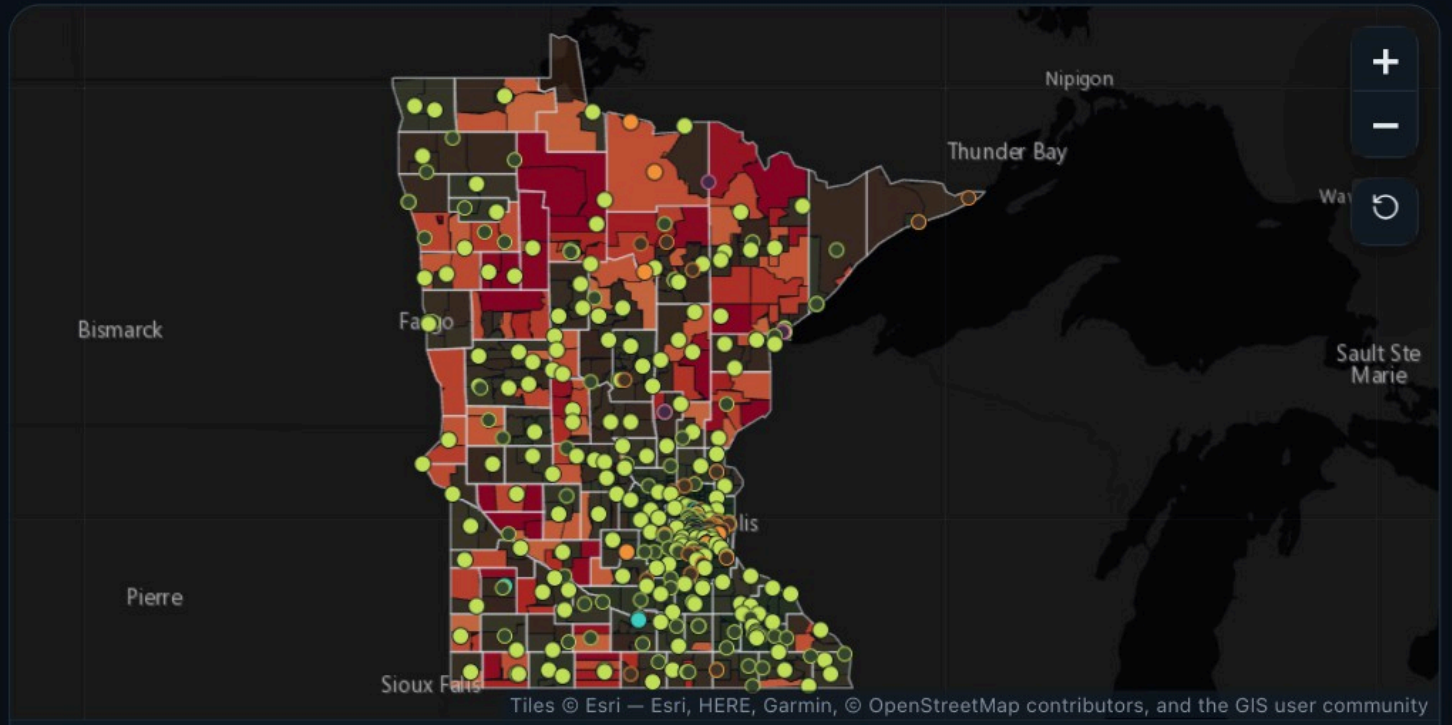
Method: five pillars, each 0–1 from public data: need (N 35 · V 15) × access (S 15 · T 20 · M 15), multiplied into a 0–100 score · **Source:** mi.fei: 1,501 Census 2020 tracts, 2...

Open Equity index (FEI) →

Daniela

Barriers concentrate in the rural west and north, far from where most food shelves are.

✓ Barrier score ✓ + Food shelves 3 Highest-barrier fifth



FEI

Fewer barriers — More barriers

Brick & mortar · Outreach / mobile · Tribal · Campus

Only the highest-barrier fifth of tracts is lit. Start new-shelf and mobile-stop planning here.

HIGHEST-BARRIER COUNTIES (FEI, 0–100)

Mahnomen	70
Clearwater	70
Swift	69
Wilkin no shelf	65
Beltrami	63
Nobles	63

RESIDENTS IN THE HIGHEST-BARRIER FIFTH

1.1M

OF THEM, CHILDREN

250.2K

Monday · Siting: start new-shelf and mobile-stop planning in the highest-barrier tracts, not where visits are highest.

Method: FEI checked against data it never used: rank correlation with SNAP use and Feeding America's rate, county bootstrap · **Source:** fei_tracts.csv, fei_validation.csv; si...

Open Equity index map →

Q6 GRADUATE ADDENDUM: A NEED MEASURE THAT IS NOT DEMAND

Daniela

Need is a household without reliable food, whether or not it asks, and here is the evidence that would prove our measure wrong.

✓ 1–2 Definition · why need | 2 3–4 Reject · revise

1 Definition, stated plainly

"A household lacks reliable resources for enough food over a set period, including people who never come to a shelf."

Higher hardship and transport barriers should mark more of it, among users and non-users alike.

2 Why it measures need, not demand

- ✓ Feeding America food-insecurity rate
- ✓ Census: share in poverty
- ✓ Census: households without a car

- ✗ ~~Visits (HouseholdsReg)~~
- ✗ ~~Pounds (PoundsReg)~~
- ✗ ~~Individuals (IndividualsReg)~~

Only public hardship data go in, so the score is free to disagree with demand.

3 Evidence that would make us reject it

A household food-security survey that includes non-users finds **no more hardship in high-score counties**, with a bound that rules out a useful difference. Not yet run: we claim no survey validation.

4 Evidence that would make us revise it

Dropping one input or changing the Feeding America edition moves a county **by more than a fifth** of the ranking; or fieldwork shows the car measure mostly flags well-served urban households.

CHECKS RUN NOW

✓ **Passed:** higher where more people use SNAP (agreement 0.66; the rule rejects at zero or below)

✓ **Only just:** with poverty held fixed, 0.28 (range 0.02 to 0.50)

Monday · Measurement: use the score to decide where to look, not how much food to move.

Method: Q6 score = mean county percentile of three public inputs; checks are rank correlations over 82 SNAP units, bootstrap 95% ranges · **Source:** docs/reference/q6_a...

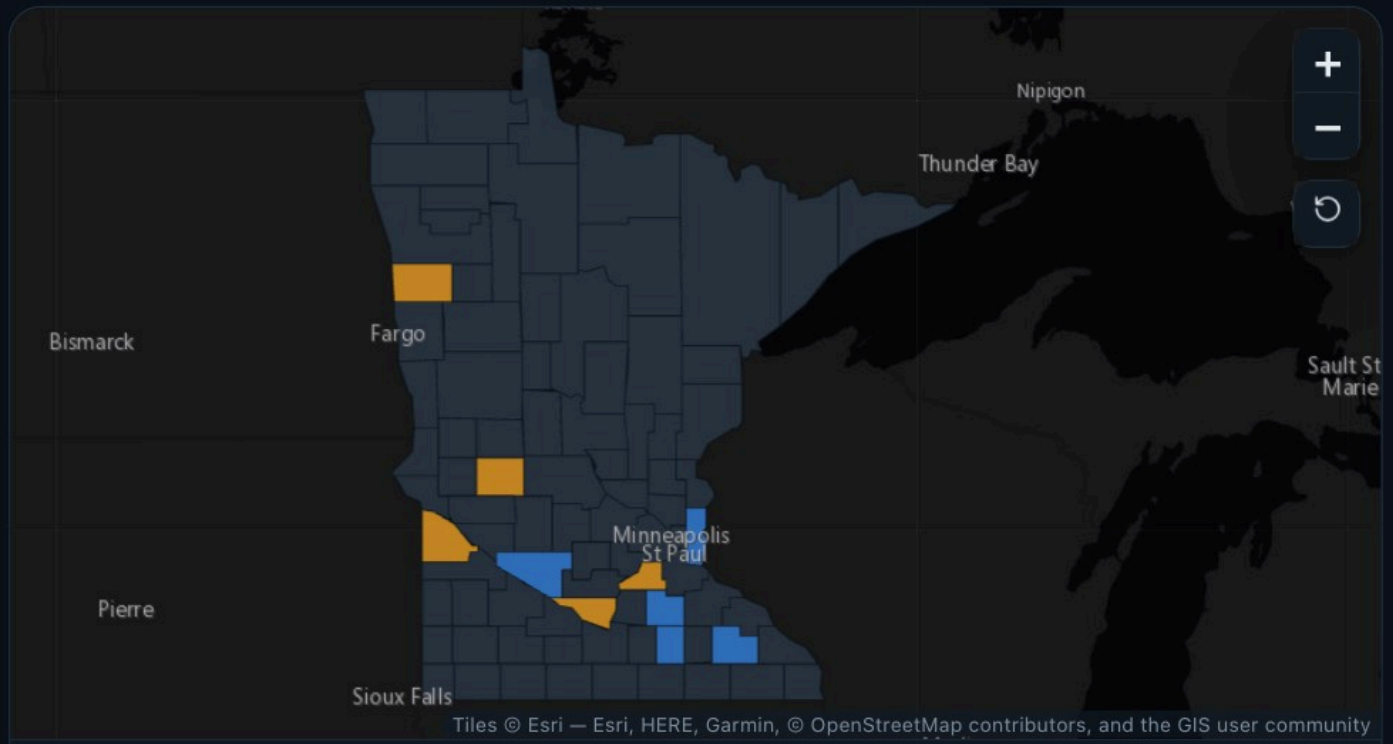
Open Equity index (FEI) →

Q1 NEED ESTIMATES VS OBSERVED ACTIVITY

Daniela

Busy counties are just big counties. Per resident, visits, pounds and individuals don't follow Feeding America's need estimate.

✓ Which measure tracks need? ✓ Per resident? ✓ Which counties break it? 4 Why?



Less activity than need suggests More activity than need suggests

Top five each way, visits per resident vs Feeding America's rate, 2024.

WHAT GOES WITH MORE OR LESS ACTIVITY PER RESIDENT

✓ **Distance to a shelf:** farther away, fewer visits per resident
rank correlation -0.33 visits · -0.41 individuals · -0.21 pounds

✓ **Households without a car:** more of them, more activity per resident (shelves within walking or transit reach)
+0.19 SD in a model with poverty, rurality and site density

– **Poverty, rurality, SNAP use:** no clear link once the others are held fixed
poverty -0.15 to 0.26; rural -0.20 to 0.12

County traits explain only about 13% of the differences. Local reasons (one dominant shelf, where it is booked, hubs) matter more.

Monday · Allocation: allocating by visits allocates by county size; audit the outlier counties before reading them as unmet need.

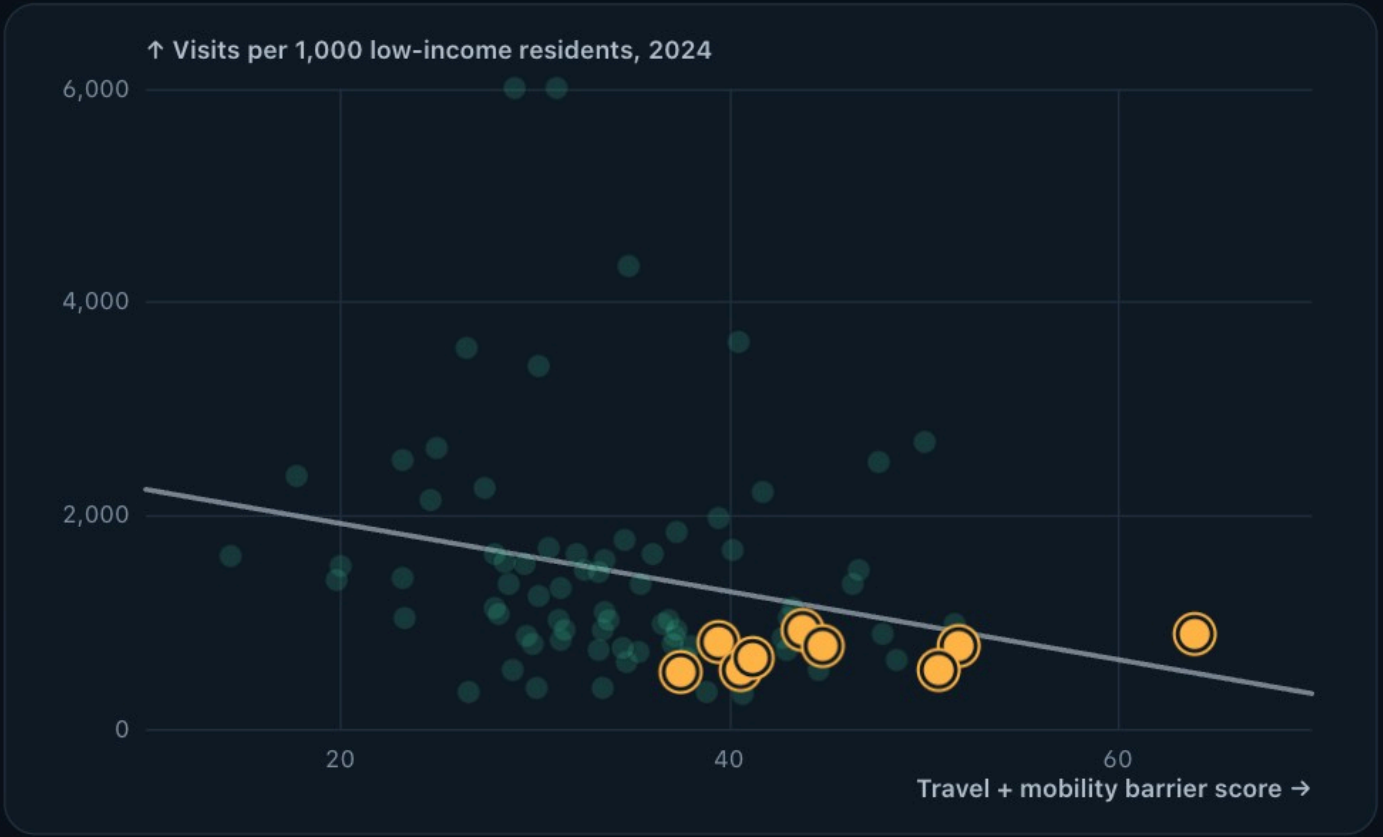
Method: Spearman rank correlation, 2024 activity vs Map the Meal Gap 2026 edition (2024 data), county bootstrap; outliers = largest residuals of activity per resident on the r...

Open Map explorer →

Daniela

Where it is harder to get to a shelf, fewer low-income residents show up. That is need not turning into demand.

✓ Barriers vs the three measures | 2 Where they disagree



9 COUNTIES: TOP FIFTH ON BARRIERS AND ON OUR Q6 NEED LIST

Aitkin | Beltrami | Chippewa | Clearwater | Lyon | Nobles | Swift | Todd | Wadena

A call list for local checks: hours, transport, language, unreported shelves.

Not proof of unmet need: the Q6 screen on its own flags 14 counties, about what random pairing gives (~16).

Monday · Measurement: call these counties first about hours, transport and language before moving supply; a flag is a call, not a verdict.

Method: Spearman rank correlation, travel and mobility pillars vs 2024 activity per 1,000 residents below 200% of poverty; Q6 chance comparison by random pairin...

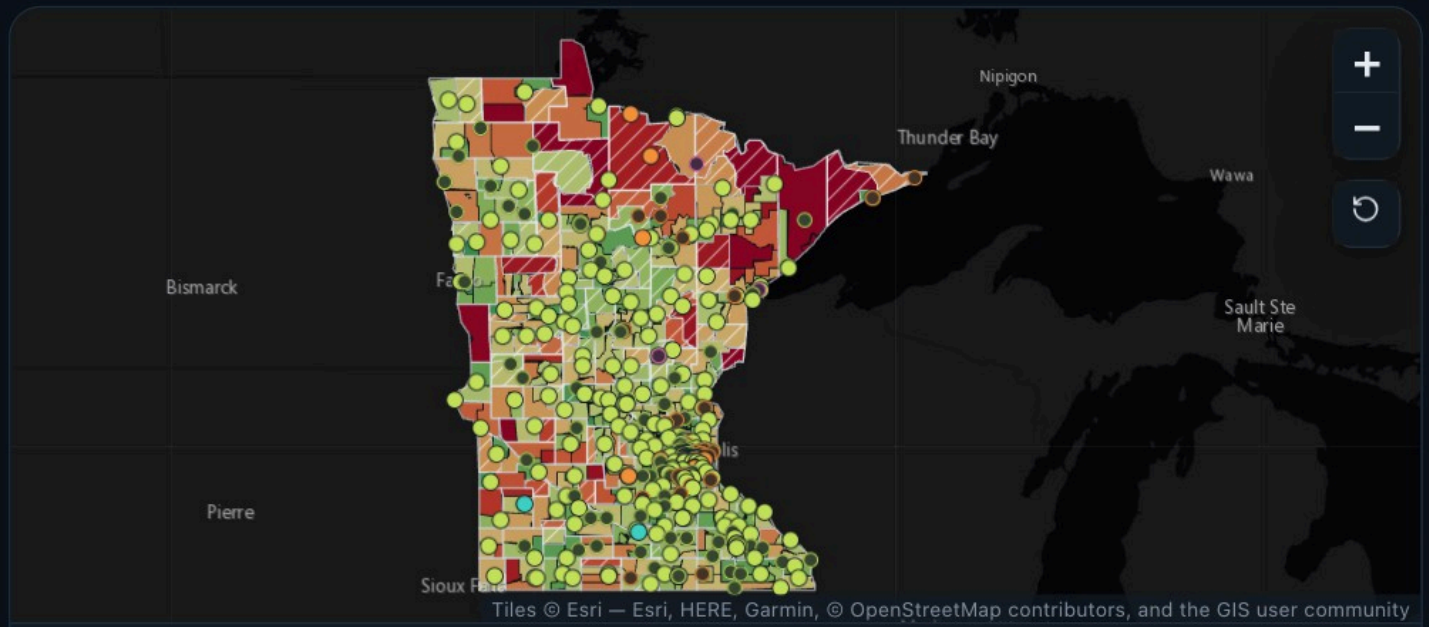
Open Map explorer: gap layer →

Q8 LOW-INCOME, LOW-ACCESS TRACTS AND NETWORK REACH

Daniela

Food shelves repeat the grocery gap instead of filling it: low-access tracts are less likely to have a shelf, and farther from one.

✓ Low-income, low-access tracts ✓ Do shelves reach them? 3 Fill or repeat the gap?



MILES TO NEAREST SHELF

0 4 8 11 15

Closer Farther

Low-income, low-access

Miles to the nearest shelf. Hatched = low-income, low-access.

AVERAGE MILES TO THE NEAREST SHELF (STRAIGHT LINE · BY ROAD)

In the country

Low access	7.9 mi
Better access	4.1 mi

By road: 10.5 vs 5.4 mi

In town

Low access	1.4 mi
Better access	0.9 mi

LIMITS

USDA's LILA is 2019 data on 2010 tracts, carried to 2020 tracts by area. Miles are to the nearest reporting shelf, not travel time or hours. A shelf inside a tract is a footprint, not proof residents are served.

Monday · Siting: check catchments, hours and mobile stops in low-access tracts with no shelf before opening anything new.

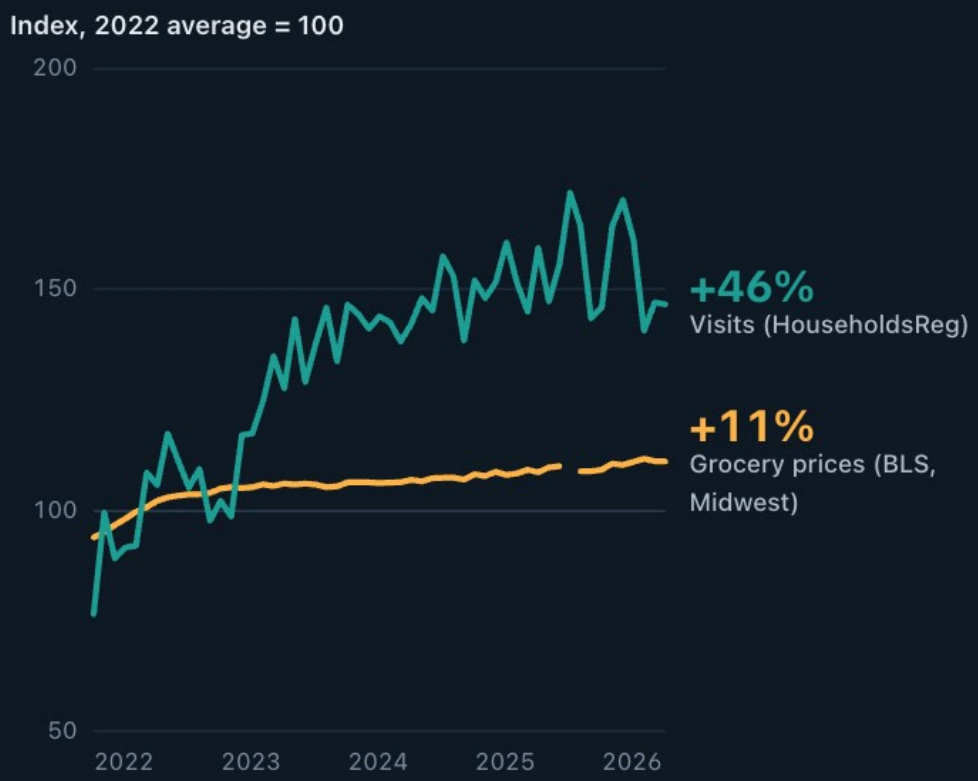
Method: low-income tracts split by USDA access flag, urban and rural compared separately; population-weighted straight-line and road miles from block points ·...

Open Map explorer: LILA overlay →

Daniela

Grocery prices are a weak early warning: only individuals served follow them, a month later. The calendar matters more.

✓ Do prices lead activity? | 2 What cycles remain?



1 Do more people come when prices rise? Right away or later?

Only individuals served, about +3% a month later; visits and pounds show no clear change.

2 What cycles remain after prices?

Each extra working day in a month adds



Slower growth than January in April and September (visits) and December and February (individuals). Net of working days, August is an ordinary month.

Monday · Allocation: warn shelves on the calendar and recent trend; don't promise donors a county surge from prices alone.

Method: monthly log changes on lags 0–6 chosen by AICc, with season, working days and the April 2023 pulse; HAC 95% ranges; working-day elasticity wit...

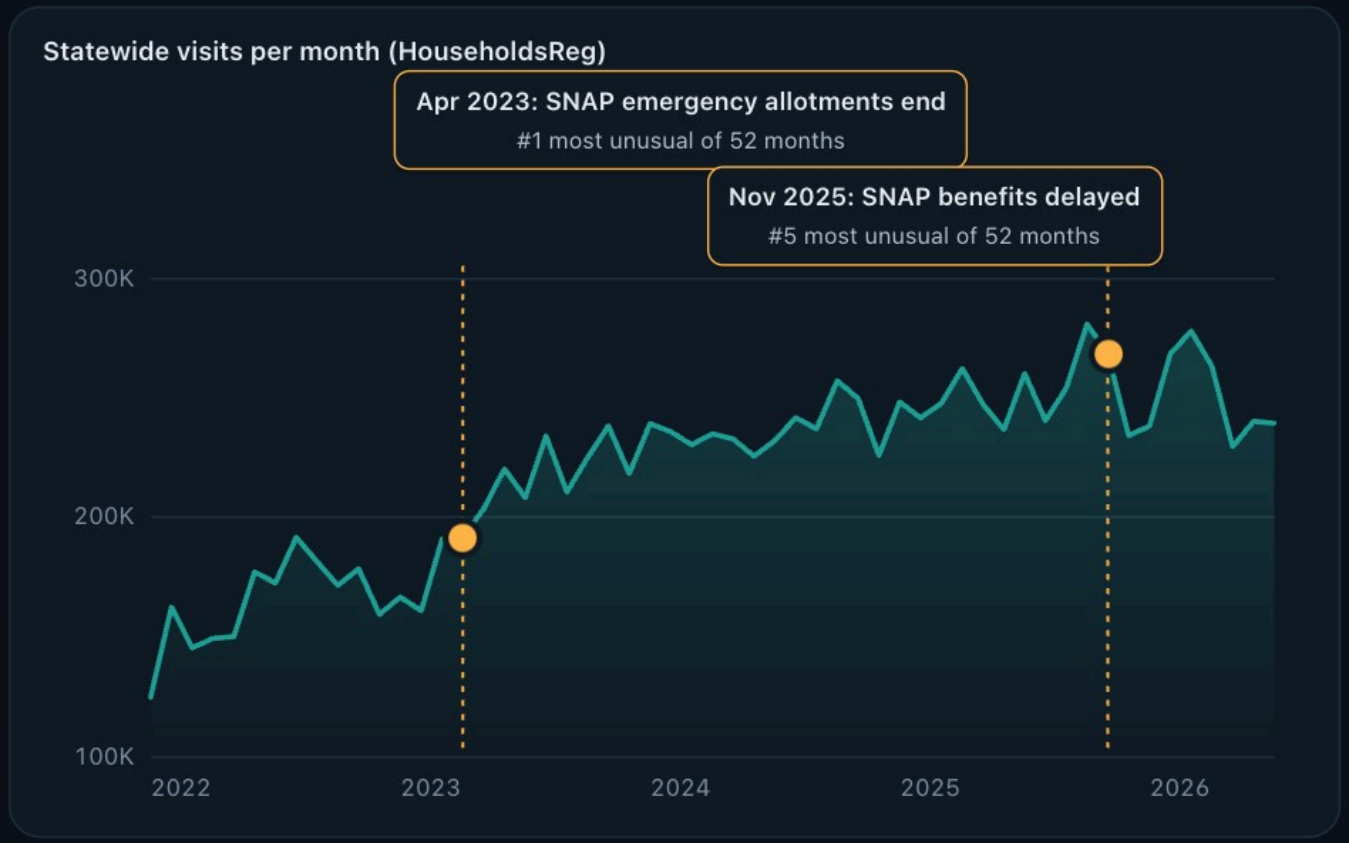
Open Command center: policy watch →

Q5 SNAP ENROLLMENT AND FOOD-SHELF USE

Daniela

SNAP and food shelves move together: more enrollment in a county comes with more visits, not fewer.

✓ Together or substitutes? ✓ When benefits change 3 Does it differ by place?



1 Together, or does one replace the other?

Together: +0.27% visits per +1% SNAP enrollment in the county.

2 What happens when benefits change?

Both SNAP shocks fell in unusually high months for visits.

3 Does it differ by place?

No clear difference for rural counties (visits -0.11, range -0.73 to 0.51), and November 2025 hit high-SNAP counties no harder. Tribal SNAP agencies are reported separately.

Monday · Policy: tell lawmakers SNAP and food shelves rise and fall together; ask DCYF for advance notice of benefit gaps so shelves can stock up.

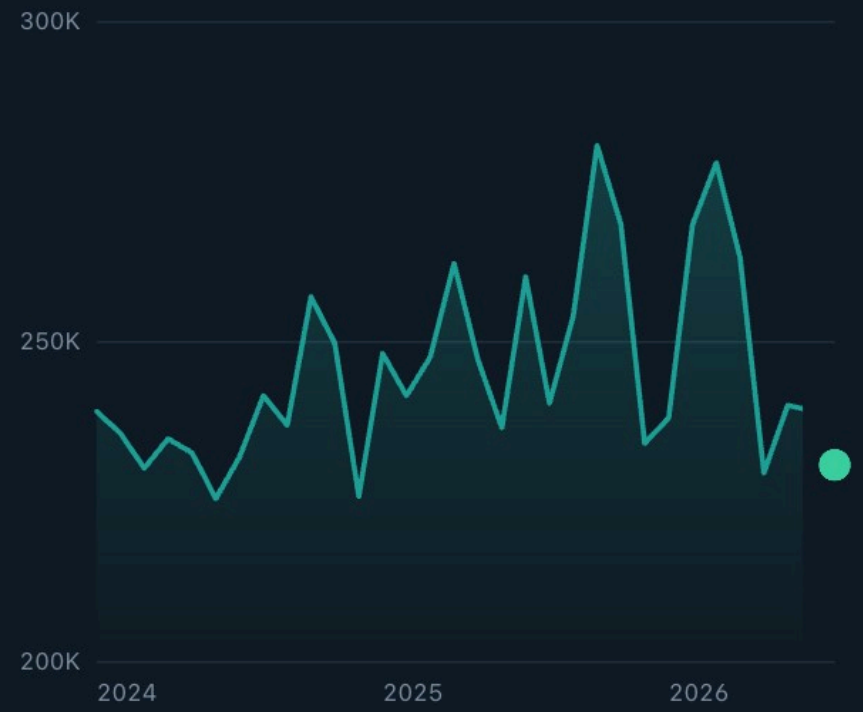
Method: county and month fixed effects with site count and unemployment, clustered by SNAP unit; benefit events ranked among 52 months after season; ...

Open Command center: policy watch →

August: about 230.6K visits and 12.1M pounds, with county error about 6× smaller than "repeat last August".

THE RULE Visits per working day **11,469** May ×1 **10,906** Jun ×2 **10,870** Jul ×3 → weighted pace **10,982 a day** × **21 working days**, in every county and field

Statewide visits per month July 239,143 → August 230,620 (-3.6%)



WHAT WE SUBMITTED (SUM OF 87 COUNTIES) AND ITS VALIDATION SCORE

Field	Submitted	Error, our rule	Error, ML model	Repeat last Aug.
Visits HouseholdsReg	230,620	371	654	2,295
Pounds distributed PoundsReg	12,122,052	42,646	42,813	85,367
Children ChildrenReg	251,086	584	620	2,570
Adults AdultsReg	378,592	653	572	3,442
Seniors SeniorsReg	82,965	234	242	727

Error: root mean squared county error over the last three Augusts, the scoring metric; lower is better. A gradient-boosting ML model beat our rule only on adults, so we submitted the rule.

Graduate addendum: children + adults + seniors add up to individuals in every county. Wilkin = 0 (no reporting shelf).

Monday · Allocation: set August orders from each county's forecast; the shelf portal shows the same number to the shelf.

Method: recent3_weighted (D-045), applied per county and field. Error: root mean squared county error over the August 2023–2025 folds (the scoring metric...

Open Map explorer: forecast layer →

Q1 Q6 Q7 Q8 FOODBRIDGE, LIVE

Roshit Jacob

The same evidence, in the tool staff would open on Monday morning.

1 Command center Q1 · Q7

Click a county: its 24-month trend, the August forecast, its sites by type and open alerts.

Roshit · 12 s Open →

2 Food shelf portal Q7

The monthly report in the network's own fields, and next month's forecast for the shelf itself.

Roshit · 8 s Open →

3 Site planner Q8 · Q6

Place a shelf in a high-barrier county: residents brought within reach and the driving time.

Jacob · 12 s Open →

4 Supply allocation Q1 · Q6

Split a supply pool by need and barriers instead of by visits alone, and see which counties gain.

Jacob · 8 s Open →

Monday · All four: managers allocate and site from one screen; each shelf sees its own forecast and reports in the same fields.

Source: county and tract aggregates only; forecasts labelled; the AI briefing retrieves and cites figures and never computes one. Browser Back returns here.

Open Command center →

Q1 Q4 Q5 Q6 Q7 Q8 WHAT CHANGES ON MONDAY

Jacob

The Food Group can't see need directly. It can see where barriers keep need from becoming demand, and act on it.

Q1 · Q6 · Q7

Allocate

Split supply by barriers as well as visits; order August on each county's forecast.

Q8 · Q6

Site

Check the 9 priority counties and shelf-less low-access tracts first; test a site in the planner.

Q5 · Q4

Argue

SNAP and shelves move together: protect SNAP, and ask DCYF for advance notice of benefit gaps.

Q6

Measure

Report barriers next to visits, so results are judged on who still can't reach food.

Thank you. Questions?

Source: FEI and the Q6 score are screening measures, not counts of unmet people; nothing here is a causal effect; every number is reproduced from code in the team repository.

Q6 APPENDIX · Q6 GRADUATE ADDENDUM

Daniela

A need definition that could be proven wrong.

1 · DEFINITION

Need: a household lacks reliable resources for adequate food over a set recall period, whether or not it asks for charitable food. Higher hardship and transport constraints should mark more of it, including among non-users.

2 · WHY NEED, NOT DEMAND

No input uses visits, pounds or individuals served. The score is built only from public hardship and access data, so it can disagree with activity, and the gap is the finding.

3 · REJECT IF

A household food-security survey that includes non-users finds no higher hardship in high-score counties, with a confidence bound that rules out a useful difference. Not yet run; we claim no survey validation.

4 · REVISE IF

Dropping one input or changing the Map the Meal Gap edition moves a county more than one fifth; or fieldwork shows the vehicle measure mostly flags well-served urban households.

CHECKS RUN NOW (82 SNAP UNITS)

need score vs snap participation: **ρ 0.66** [0.50, 0.79] · reject index if upper 95% bound <= 0

poverty input alone vs snap participation: **ρ 0.62** [0.45, 0.76] · context: one index input alone

need score vs snap participation given poverty: **ρ 0.28** [0.02, 0.50] · context: rank partial correlation holding poverty fixed

Source: docs/reference/q6_addendum.md; q6_falsification_checks.csv. Q6 score = mean county percentile of the Map the Meal Gap rate, ACS poverty and no-vehicle share; FEI is the tract-level extension.

The submitted run rate beats every simple baseline in all six fields.

Field	Repeat last August	Last 12-month average	Repeat July	3-month working-day rate	Same, weighted 1:2:3 (submitted)	Gradient-boosting ML (tested, not submitted)	80% band coverage, Aug 2025
Visits HouseholdsReg	2,295	1,450	983	424	**371**	654	91%
Pounds distributed PoundsReg	85,367	68,702	59,911	43,346	**42,646**	42,813	81%
Children ChildrenReg	2,570	2,047	1,200	695	**584**	620	81%
Adults AdultsReg	3,442	2,462	1,640	760	**653**	572	91%
Seniors SeniorsReg	727	418	281	276	**234**	242	86%
Individuals served IndividualsReg	6,383	4,786	2,921	1,403	**1,138**	915	87%

Source: Root mean squared county error over the August 2023/2024/2025 folds (q7_backtest.csv), sponsor units. Gradient-boosting ML: one fixed configuration on the same folds and counties (mi.forecas...

Jacob

FEI agrees with need signals it never used, and its ranking survives re-weighting.

Check	ρ [95%]
SNAP participation (not an input)	**0.59** [0.40, 0.73]
Q6 need score	**0.83** [0.75, 0.88]
Map the Meal Gap food-insecurity rate	**0.85** [0.76, 0.90]
Travel+mobility vs visits per low-income resident	**-0.35** [-0.52, -0.13]
Travel+mobility vs pounds per low-income resident	**-0.23** [-0.42, -0.02]
Travel+mobility vs individuals per low-income resident	**-0.32** [-0.50, -0.11]

Weighting variant	County ρ	Top fifth kept
equal	0.98	94%
need heavy	0.98	100%
access heavy	0.97	94%
no race	0.99	100%
previous additive index	0.98	100%
demand total population	0.99	100%
supply shelf count	0.82	83%
supply households	0.99	100%
decay x0.5	0.99	100%
decay x1.5	0.99	100%
network 2025	1.00	100%
location block points only	1.00	100%
road distance	1.00	100%

Source: fei_validation.csv (bootstrap Spearman ρ, 95% intervals; the Map the Meal Gap comparison is published as ρ only, D-050) and fei_sensitivity.csv.

Daniela

Provided data, plus public sources beyond them, each with publisher, grain and vintage in our catalog.

PROVIDED BY THE CHALLENGE

- The Food Group: org_FSStats and Organization extracts, January 2022–July 2026
- Feeding America Map the Meal Gap, 2024–2026 editions (licensed; cited, not redistributed)
- BLS CPI food at home: US, Midwest, Minneapolis–St. Paul
- Minnesota DCYF SNAP caseload and benefits, monthly by agency
- Minnesota Department of Education school-meal claims (context)

ADDITIONAL PUBLIC SOURCES

- U.S. Census Bureau ACS five-year 2020–2024 and 2019–2023 (county need signals, with margins of error)
- U.S. Census Bureau 2020 Census blocks, tract relationship file, cartographic boundaries, Census Geocoder
- USDA ERS Food Access Research Atlas 2019 (LILA), SNAP Retailer Access Map 2025, Rural-Urban Continuum Codes 2023
- BLS Local Area Unemployment Statistics, county monthly
- USDA FNS SNAP state data tables (cross-check)
- CDC/ATSDR Social Vulnerability Index 2022
- HUD Fair Market Rents FY2022–FY2026; Census Population Estimates Vintage 2025
- FEMA disaster declarations; EIA Minnesota gasoline prices
- OpenStreetMap (Geofabrik) road network routed with OSRM (driving access)
- Urban Institute Well-Being and Basic Needs Survey; USDA ERS and DCYF reports for definitions and policy dates

Source: docs/sources/catalog.md: one row per source (publisher, exact series, geography, time grain, vintage, URL, access date, why chosen).